

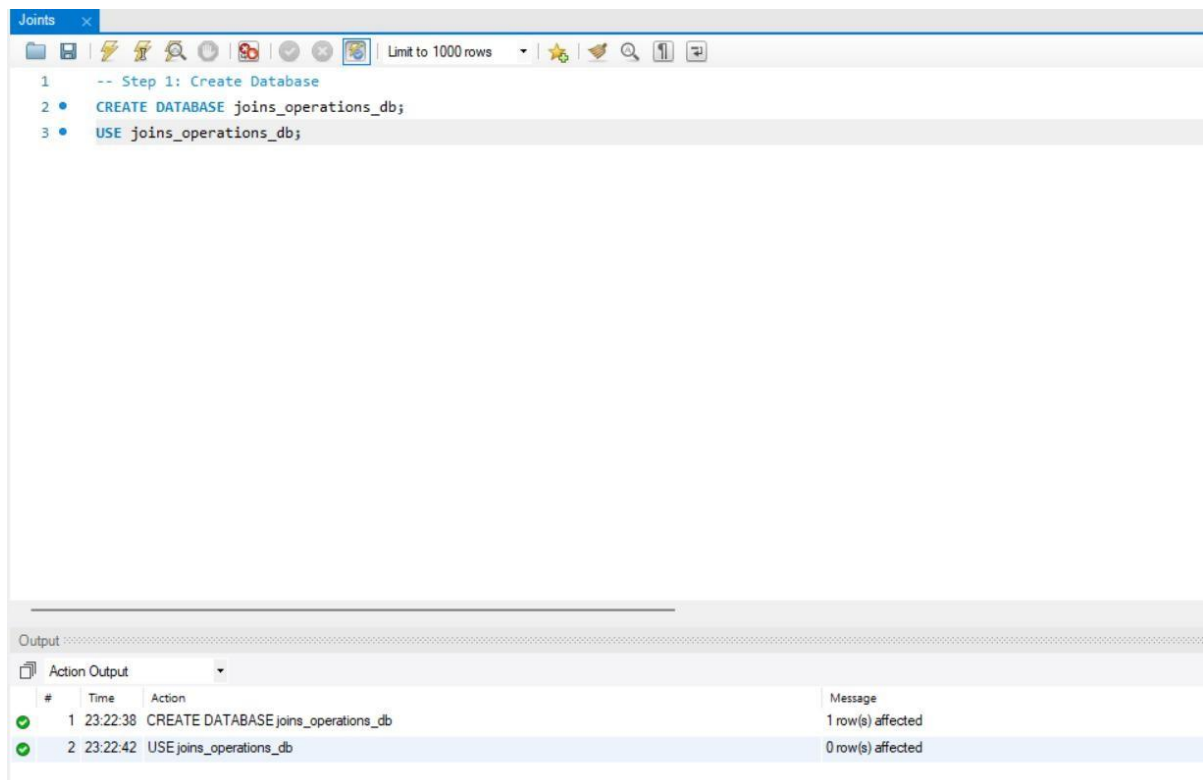
JOINS

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Question :

Perform SQL JOIN and SET operations on student and address data using CROSS JOIN, INNER JOIN, NATURAL JOIN, LEFT OUTER JOIN, RIGHT OUTER JOIN, FULL OUTER JOIN, UNION, UNION ALL, INTERSECT and MINUS/EXCEPT, including filtering of unmatched records and displaying address availability using CASE

Step 1 : Create Database



The screenshot shows a SQL IDE window titled "Joins". The main editor area contains the following SQL script:

```
-- Step 1: Create Database
1  CREATE DATABASE joins_operations_db;
2  USE joins_operations_db;
```

The "Output" pane at the bottom shows the execution results:

#	Time	Action	Message
1	23:22:38	CREATE DATABASE joins_operations_db	1 row(s) affected
2	23:22:42	USE joins_operations_db	0 row(s) affected

Joins

```

1  -- Step 1: Create Database
2  • CREATE DATABASE joins_operations_db;
3  • USE joins_operations_db;
4  -- Step 2: Create Tables
5  • CREATE TABLE class(
6      id INT,
7      name VARCHAR(30)
8  );
9
10 • CREATE TABLE class_info(
11     id INT,
12     address VARCHAR(30)
13 );

```

Output

Action Output

#	Time	Action	Message
✓ 1	23:22:38	CREATE DATABASE joins_operations_db	1 row(s) affected
✓ 2	23:22:42	USE joins_operations_db	0 row(s) affected
✓ 3	23:23:15	CREATE TABLE class(id INT, name VARCHAR(30))	0 row(s) affected
✓ 4	23:23:18	CREATE TABLE class_info(id INT, address VARCHAR(30))	0 row(s) affected

JoinsSQL

```

10 • CREATE TABLE class_info(
11     id INT,
12     address VARCHAR(30)
13 );
14 -- Step 3: Insert Data
15 • INSERT INTO class VALUES
16 (1,'Angela'),
17 (2,'Smija'),
18 (3,'Riya'),
19 (4,'Tianne'),
20 (5,'Neha');
21
22 • INSERT INTO class_info VALUES
23 (1,'KERALA'),
24 (2,'MUMBAI'),
25 (3,'CHENNAI'),
26 (7,'DELHI'),
27 (8,'GUJARAT');
28
29

```

Output

Action Output

#	Time	Action	Message
✗ 1	13:54:03	CREATE DATABASE joins_operations_db	Error Code: 1007. Can't create database 'joins_operations_db'; database exists
✗ 2	13:54:08	CREATE DATABASE joins_operations_db	Error Code: 1007. Can't create database 'joins_operations_db'; database exists
✓ 3	13:54:29	USE joins_operations_db	0 row(s) affected
✗ 4	13:54:36	CREATE TABLE class_info(id INT, address VARCHAR(30))	Error Code: 1050. Table 'class_info' already exists
✓ 5	13:54:47	INSERT INTO class VALUES (1,'Angela'), (2,'Smija'), (3,'Riya'), (4,'Tianne'), (5,'Neha')	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0

Step 2 : CROSS JOIN

JoinsSQL

```
24 (2,'MUMBAI'),
25 (3,'CHENNAI'),
26 (7,'DELHI'),
27 (8,'GUJARAT');
28 -- Step 4: CROSS JOIN
29 • SELECT *
30 FROM class
31 CROSS JOIN class_info;
32
```

Result Grid

	id	name	id	address
1	Angela	8	GUJARAT	
1	Angela	7	DELHI	
1	Angela	3	CHENNAI	
1	Angela	2	MUMBAI	
1	Angela	1	KERALA	
2	Smija	8	GUJARAT	
2	Smija	7	DELHI	
2	Smija	3	CHENNAI	
2	Smija	2	MUMBAI	
2	Smija	1	KERALA	

Result 1 x

Output

Action Output

#	Time	Action	Message
1	13:54:03	CREATE DATABASE joins_operations_db	Error Code: 1007. Can't create database 'joins_operations_db'; database exists
2	13:54:08	CREATE DATABASE joins_operations_db	Error Code: 1007. Can't create database 'joins_operations_db'; database exists
3	13:54:29	USE joins_operations_db	0 row(s) affected
4	13:54:36	CREATE TABLE class_info(id INT, address VARCHAR(30))	Error Code: 1050. Table 'class_info' already exists
5	13:54:47	INSERT INTO class VALUES (1,'Angela'), (2,'Smija'), (3,'Riya'), (4,'Tianne'), (5,'Neha')	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
6	13:55:47	SELECT * FROM class CROSS JOIN class_info LIMIT 0, 1000	50 row(s) returned

Step 3 : INNER JOIN

JoinsSQL

```
29 • SELECT *
30 FROM class
31 CROSS JOIN class_info;
32 -- Step 5: INNER JOIN
33 • SELECT *
34 FROM class
35 INNER JOIN class_info
36 ON class.id = class_info.id;
37
```

Result Grid

	id	name	id	address
1	Angela	1	KERALA	
2	Smija	2	MUMBAI	
3	Riya	3	CHENNAI	
1	Angela	1	KERALA	
2	Smija	2	MUMBAI	
3	Riya	3	CHENNAI	

Result 2 x

Output

Action Output

#	Time	Action	Message
2	13:54:08	CREATE DATABASE joins_operations_db	Error Code: 1007. Can't create database 'joins_operations_db'; database exists
3	13:54:29	USE joins_operations_db	0 row(s) affected
4	13:54:36	CREATE TABLE class_info(id INT, address VARCHAR(30))	Error Code: 1050. Table 'class_info' already exists
5	13:54:47	INSERT INTO class VALUES (1,'Angela'), (2,'Smija'), (3,'Riya'), (4,'Tianne'), (5,'Neha')	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
6	13:55:47	SELECT * FROM class CROSS JOIN class_info LIMIT 0, 1000	50 row(s) returned
7	13:56:23	SELECT * FROM class INNER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	6 row(s) returned

JoinsSQL*

```

34 FROM class
35 INNER JOIN class_info
36 ON class.id = class_info.id;
37 -- Step 6: INNER JOIN Selected Columns
38 • SELECT class.name,
39       class_info.address
40 FROM class
41 INNER JOIN class_info
42 ON class.id = class_info.id

```

Result Grid

	name	address
▶	Angela	KERALA
	Smija	MUMBAI
	Riya	CHENNAI
	Angela	KERALA
	Smija	MUMBAI
	Riya	CHENNAI

Result 3 x

Output

Action Output

#	Time	Action	Message
3	13:54:29	USE joins_operations_db	0 row(s) affected
4	13:54:36	CREATE TABLE class_info(id INT, address VARCHAR(30))	Error Code: 1050. Table 'class_info' already exists
5	13:54:47	INSERT INTO class VALUES (1,'Angela'), (2,'Smija'), (3,'Riya'), (4,'Tianne'), (5,'Neha')	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
6	13:55:47	SELECT * FROM class CROSS JOIN class_info LIMIT 0, 1000	50 row(s) returned
7	13:56:23	SELECT * FROM class INNER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	6 row(s) returned
8	13:56:52	SELECT class.name, class_info.address FROM class INNER JOIN class_info ON class.id = class_info.id L...	6 row(s) returned

Step 4 : NATURAL JOIN

JoinsSQL*

```

31 CROSS JOIN class_info;
32 -- Step 5: INNER JOIN
33 • SELECT *
34 FROM class
35 INNER JOIN class_info
36 ON class.id = class_info.id;
37 -- Step 6: INNER JOIN Selected Columns
38 • SELECT class.name,
39       class_info.address
40 FROM class
41 INNER JOIN class_info
42 ON class.id = class_info.id
43 -- Step 7: NATURAL JOIN
44 ✖ SELECT *
45 FROM class
46 NATURAL JOIN class_info;
47
48
49
50

```

Output

Action Output

#	Time	Action	Message
4	13:54:36	CREATE TABLE class_info(id INT, address VARCHAR(30))	Error Code: 1050. Table 'class_info' already exists
5	13:54:47	INSERT INTO class VALUES (1,'Angela'), (2,'Smija'), (3,'Riya'), (4,'Tianne'), (5,'Neha')	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
6	13:55:47	SELECT * FROM class CROSS JOIN class_info LIMIT 0, 1000	50 row(s) returned
7	13:56:23	SELECT * FROM class INNER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	6 row(s) returned
8	13:56:52	SELECT class.name, class_info.address FROM class INNER JOIN class_info ON class.id = class_info.id L...	6 row(s) returned
9	13:57:54	SELECT class.name, class_info.address FROM class INNER JOIN class_info ON class.id = class_info.id --...	Error Code: 1064. You have an error in your SQL syntax; check the manual that corres

Step 5 : LEFT OUTER JOIN

JoinsSQL*

```
44 SELECT *
45 FROM class
46 NATURAL JOIN class_info;
47 -- Step 8: LEFT OUTER JOIN
48 SELECT *
49 FROM class
50 LEFT OUTER JOIN class_info
51 ON class.id = class_info.id;
52
```

Limit to 1000 rows

Result Grid

	id	name	id	address
1	Angela	1	KERALA	
2	Smija	2	MUMBAI	
3	Riya	3	CHENNAI	
4	Tianne	NULL	NULL	
5	Neha	NULL	NULL	

Result 4 x

Output

Action Output

#	Time	Action	Message
5	13:54:47	INSERT INTO class VALUES (1,'Angela'), (2,'Smija'), (3,'Riya'), (4,'Tianne'), (5,'Neha')	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
6	13:55:47	SELECT * FROM class CROSS JOIN class_info LIMIT 0, 1000	50 row(s) returned
7	13:56:23	SELECT * FROM class INNER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	6 row(s) returned
8	13:56:52	SELECT class.name, class_info.address FROM class INNER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	6 row(s) returned
9	13:57:54	SELECT class.name, class_info.address FROM class INNER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version 5.7.26
10	13:58:37	SELECT * FROM class LEFT OUTER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	10 row(s) returned

Automated disabled. Manually current c toggle

JoinsSQL*

```
49 FROM class
50 LEFT OUTER JOIN class_info
51 ON class.id = class_info.id;
52 -- Step 9: LEFT JOIN with NULL
53 SELECT *
54 FROM class
55 LEFT JOIN class_info
56 ON class.id = class_info.id
57 WHERE class_info.id IS NULL;
```

Limit to 1000 rows

Result Grid

	id	name	id	address
4	Tianne	NULL	NULL	
5	Neha	NULL	NULL	
4	Tianne	NULL	NULL	
5	Neha	NULL	NULL	

Result 5 x

Output

Action Output

#	Time	Action	Message
6	13:55:47	SELECT * FROM class CROSS JOIN class_info LIMIT 0, 1000	50 row(s) returned
7	13:56:23	SELECT * FROM class INNER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	6 row(s) returned
8	13:56:52	SELECT class.name, class_info.address FROM class INNER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	6 row(s) returned
9	13:57:54	SELECT class.name, class_info.address FROM class INNER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version 5.7.26
10	13:58:37	SELECT * FROM class LEFT OUTER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	10 row(s) returned
11	13:59:03	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id WHERE class_info.id IS NULL LIMIT 0, 1000	4 row(s) returned

Automated disabled. Manually current c toggle

Step 6 : RIGHT OUTER JOIN

JoinsSQL™

```
61 RIGHT OUTER JOIN class_info
62 ON class.id = class_info.id;
63
64 -- Step 11: RIGHT JOIN with NULL
65 • SELECT *
66 FROM class
67 RIGHT JOIN class_info
68 ON class.id = class_info.id
69 WHERE class.id IS NULL;_
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [↗](#)

	id	name	id	address
▶	NULL	NULL	7	DELHI
	NULL	NULL	8	GUJARAT

Result 7

Output

Action Output

#	Time	Action	Message
✓ 8	13:56:52	SELECT class.name, class_info.address FROM class INNER JOIN class_info ON class.id = class_info.id L...	6 row(s) returned
✗ 9	13:57:54	SELECT class.name, class_info.address FROM class INNER JOIN class_info ON class.id = class_info.id ~...	Error Code: 1064. You have an error in your SQL syntax; check the manual
✓ 10	13:58:37	SELECT * FROM class LEFT OUTER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	10 row(s) returned
✓ 11	13:59:03	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id WHERE class_info.id IS NULL LIMIT...	4 row(s) returned
✓ 12	13:59:43	SELECT * FROM class RIGHT OUTER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	8 row(s) returned
✓ 13	14:00:27	SELECT * FROM class RIGHT JOIN class_info ON class.id = class_info.id WHERE class.id IS NULL LIMIT 0, ...	2 row(s) returned

JOINS

```
60 FROM class
61 RIGHT OUTER JOIN class_info
62 ON class.id = class_info.id;
63 -- Step 11: RIGHT JOIN with NULL
64 • SELECT *
65 FROM class
66 RIGHT JOIN class_info
67 ON class.id = class_info.id
68 WHERE class.id IS NULL;|
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [↗](#)

	id	name	id	address
▶	NULL	NULL	7	NOIDA
	NULL	NULL	8	PANIPAT

Result 8

Output

Action Output

#	Time	Action	Message
✓ 9	23:27:39	SELECT class.name, class_info.address FROM class INNER JOIN class_info ON class.id = class_info.id L...	3 row(s) returned
✓ 10	23:29:44	SELECT * FROM class NATURAL JOIN class_info LIMIT 0, 1000	3 row(s) returned
✓ 11	23:31:37	SELECT * FROM class LEFT OUTER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	5 row(s) returned
✓ 12	23:32:28	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id WHERE class_info.id IS NULL LIMIT...	2 row(s) returned
✓ 13	23:33:36	SELECT * FROM class RIGHT OUTER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	5 row(s) returned
✓ 14	23:34:21	SELECT * FROM class RIGHT JOIN class_info ON class.id = class_info.id WHERE class.id IS NULL LIMIT 0, ...	2 row(s) returned

Step 7: FULL OUTER JOIN

JoinsSQL* x

```
73 LEFT JOIN class_info
74 ON class.id = class_info.id
75
76 UNION
77
78 SELECT *
79 FROM class
80 RIGHT JOIN class_info
81 ON class.id = class_info.id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	id	name	id	address
▶	1	Angela	1	KERALA
	2	Smija	2	MUMBAI
	3	Riya	3	CHENNAI
	4	Tianne	NULL	NULL
	5	Neha	NULL	NULL
	7	DELHI		
	8	GUJARAT		

Result 8 x Read Only

Output

Action Output

#	Time	Action	Message
9	13:57:54	SELECT class.name, class_info.address FROM class INNER JOIN class_info ON class.id = class_info.id ~...	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds
10	13:58:37	SELECT * FROM class LEFT OUTER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	10 row(s) returned
11	13:59:03	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id WHERE class_info.id IS NULL LIMIT...	4 row(s) returned
12	13:59:43	SELECT * FROM class RIGHT OUTER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	8 row(s) returned
13	14:00:27	SELECT * FROM class RIGHT JOIN class_info ON class.id = class_info.id WHERE class.id IS NULL LIMIT 0, ...	2 row(s) returned
14	14:01:51	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id UNION SELECT * FROM class RIG...	7 row(s) returned

JoinsSQL* x

```
89
90 UNION
91
92 SELECT *
93 FROM class
94 RIGHT JOIN class_info
95 ON class.id = class_info.id
96 WHERE class.id IS NULL;
97
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	id	name	id	address
▶	4	Tianne	NULL	NULL
	5	Neha	NULL	NULL
	7	DELHI		
	8	GUJARAT		

Result 9 x Read Only Context

Output

Action Output

#	Time	Action	Message
10	13:58:37	SELECT * FROM class LEFT OUTER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	10 row(s) returned
11	13:59:03	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id WHERE class_info.id IS NULL LIMIT...	4 row(s) returned
12	13:59:43	SELECT * FROM class RIGHT OUTER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	8 row(s) returned
13	14:00:27	SELECT * FROM class RIGHT JOIN class_info ON class.id = class_info.id WHERE class.id IS NULL LIMIT 0, ...	2 row(s) returned
14	14:01:51	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id UNION SELECT * FROM class RIG...	7 row(s) returned
15	14:02:28	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id WHERE class_info.id IS NULL UNI...	4 row(s) returned

Step 8 : SET OPERATIONS

[illegible]

➤ UNION

JoinsSQL x

```
112 (3,'Smija');
113 -- Step 16: UNION
114 • SELECT *
115 FROM first_table
116
117 UNION
118
119 SELECT *
120 FROM second_table;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Contents: |

id	name
1	Angela
2	Joel
3	Smija

Result 38 x

Output

Action Output

#	Time	Action	Message
✓ 16	15:34:26	CREATE TABLE first_table(id INT, name VARCHAR(30))	0 row(s) affected
✓ 17	15:34:31	CREATE TABLE second_table(id INT, name VARCHAR(30))	0 row(s) affected
✓ 18	15:35:18	INSERT INTO first_table VALUES (1,'Angela'), (2,'Joel')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
✓ 19	15:35:21	INSERT INTO second_table VALUES (2,'Joel'), (3,'Smija')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
✓ 20	15:35:28	INSERT INTO second_table VALUES (2,'Joel'), (3,'Smija')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
✓ 21	15:35:53	SELECT * FROM first_table UNION SELECT * FROM second_table	3 row(s) returned

JoinsSQL x

```
120 FROM second_table;
121 -- Step 17: UNION Names
122 • SELECT name
123 FROM first_table
124
125 UNION
126
127 SELECT name
128 FROM second_table;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Contents: |

name
Angela
Joel
Smija

Result 39 x

Output

Action Output

#	Time	Action	Message
✓ 17	15:34:31	CREATE TABLE second_table(id INT, name VARCHAR(30))	0 row(s) affected
✓ 18	15:35:18	INSERT INTO first_table VALUES (1,'Angela'), (2,'Joel')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
✓ 19	15:35:21	INSERT INTO second_table VALUES (2,'Joel'), (3,'Smija')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
✓ 20	15:35:28	INSERT INTO second_table VALUES (2,'Joel'), (3,'Smija')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
✓ 21	15:35:53	SELECT * FROM first_table UNION SELECT * FROM second_table	3 row(s) returned
✓ 22	15:37:02	SELECT name FROM first_table UNION SELECT name FROM second_table	3 row(s) returned

➤ UNION ALL

```
JoinsSQL x
-- Step 18: UNION ALL
129
130 • SELECT *
131 FROM first_table
132
133 UNION ALL
134
135 SELECT *
136 FROM second_table;
137 -- Step 19: Count UNION ALL Records
```

Result Grid | Filter Rows: | Exports: | Wrap Cell Contents: IA

	id	name
▶	1	Angela
	2	Joel
	2	Joel
	3	Smija
	2	Joel
	3	Smija

Result 40 x

Output

Action Output

#	Time	Action	Message
✓	18 15:35:18	INSERT INTO first_table VALUES (1,'Angela'), (2,'Joel')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
✓	19 15:35:21	INSERT INTO second_table VALUES (2,'Joel'), (3,'Smija')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
✓	20 15:35:28	INSERT INTO second_table VALUES (2,'Joel'), (3,'Smija')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
✓	21 15:35:53	SELECT * FROM first_table UNION SELECT * FROM second_table	3 row(s) returned
✓	22 15:37:02	SELECT name FROM first_table UNION SELECT name FROM second_table	3 row(s) returned
✓	23 15:37:46	SELECT * FROM first_table UNION ALL SELECT * FROM second_table	6 row(s) returned

```
JoinsSQL x
135 SELECT *
136 FROM second_table;
137 -- Step 19: Count UNION ALL Records
138 • SELECT COUNT(*)
139 FROM (
140     SELECT * FROM first_table
141
142     UNION ALL
143
```

Result Grid | Filter Rows: | Exports: | Wrap Cell Contents: IA

COUNT(*)
▶ 6

Result 41 x

Output

Action Output

#	Time	Action	Message
✓	19 15:35:21	INSERT INTO second_table VALUES (2,'Joel'), (3,'Smija')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
✓	20 15:35:28	INSERT INTO second_table VALUES (2,'Joel'), (3,'Smija')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
✓	21 15:35:53	SELECT * FROM first_table UNION SELECT * FROM second_table	3 row(s) returned
✓	22 15:37:02	SELECT name FROM first_table UNION SELECT name FROM second_table	3 row(s) returned
✓	23 15:37:46	SELECT * FROM first_table UNION ALL SELECT * FROM second_table	6 row(s) returned
✓	24 15:38:28	SELECT COUNT(*) FROM (SELECT * FROM first_table UNION ALL SELECT * FROM second_table ...	1 row(s) returned

➤ INTERSECT

```
JoinsSQL x
147 • SELECT *
148 FROM first_table
149
150 INTERSECT
151 SELECT *
152 FROM second_table;_
153 -- Step 21: INTERSECT Names
154 • SELECT name
155 FROM first_table
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	id	name
▶	2	Joel

Result 42 x

Output			
Action Output			
#	Time	Action	Message
✓ 20	15:35:28	INSERT INTO second_table VALUES (2,'Joel'), (3,'Smija')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
✓ 21	15:35:53	SELECT * FROM first_table UNION SELECT * FROM second_table	3 row(s) returned
✓ 22	15:37:02	SELECT name FROM first_table UNION SELECT name FROM second_table	3 row(s) returned
✓ 23	15:37:46	SELECT * FROM first_table UNION ALL SELECT * FROM second_table	6 row(s) returned
✓ 24	15:38:28	SELECT COUNT(*) FROM (SELECT * FROM first_table UNION ALL SELECT * FROM second_table ...	1 row(s) returned
✓ 25	15:39:06	SELECT * FROM first_table INTERSECT SELECT * FROM second_table	1 row(s) returned

```
JoinsSQL x
153 -- Step 21: INTERSECT Names
154 • SELECT name
155 FROM first_table
156
157 INTERSECT
158
159 SELECT name
160 FROM second_table;_
161 -- Step 22: MINUS
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	name
▶	Joel

Result 43 x

Output			
Action Output			
#	Time	Action	Message
✓ 21	15:35:53	SELECT * FROM first_table UNION SELECT * FROM second_table	3 row(s) returned
✓ 22	15:37:02	SELECT name FROM first_table UNION SELECT name FROM second_table	3 row(s) returned
✓ 23	15:37:46	SELECT * FROM first_table UNION ALL SELECT * FROM second_table	6 row(s) returned
✓ 24	15:38:28	SELECT COUNT(*) FROM (SELECT * FROM first_table UNION ALL SELECT * FROM second_table ...	1 row(s) returned
✓ 25	15:39:06	SELECT * FROM first_table INTERSECT SELECT * FROM second_table	1 row(s) returned
✓ 26	15:39:33	SELECT name FROM first_table INTERSECT SELECT name FROM second_table	1 row(s) returned

➤ MINUS

JoinsSQL x

Limit to 1000 rows

```
161 -- Step 22: MINUS
162 • SELECT f.*
163 FROM first_table f
164 WHERE NOT EXISTS (
165     SELECT 1
166     FROM second_table s
167     WHERE s.id = f.id
168     AND s.name = f.name
169 );
```

Result Grid

	id	name
▶	1	Angela

Export: | Wrap Cell Contents: [IA](#)

first_table 44 x

Output

Action Output

#	Time	Action	Message
✓ 22	15:37:02	SELECT name FROM first_table UNION SELECT name FROM second_table	3 row(s) returned
✓ 23	15:37:46	SELECT * FROM first_table UNION ALL SELECT * FROM second_table	6 row(s) returned
✓ 24	15:38:28	SELECT COUNT(*) FROM (SELECT * FROM first_table UNION ALL SELECT * FROM second_table ...	1 row(s) returned
✓ 25	15:39:06	SELECT * FROM first_table INTERSECT SELECT * FROM second_table	1 row(s) returned
✓ 26	15:39:33	SELECT name FROM first_table INTERSECT SELECT name FROM second_table	1 row(s) returned
✓ 27	15:40:25	SELECT f.* FROM first_table f WHERE NOT EXISTS (SELECT 1 FROM second_table s WHERE s.id...	1 row(s) returned

JoinsSQL x

Limit to 1000 rows

```
169 );
170 -- Step 23: MINUS Names
171 • SELECT f.name
172 FROM first_table f
173 WHERE NOT EXISTS (
174     SELECT 1
175     FROM second_table s
176     WHERE s.name = f.name
177 );
```

Result Grid

	name
▶	Angela

Export: | Wrap Cell Content: [IA](#)

first_table 45 x

Output

Action Output

#	Time	Action	Message
✓ 23	15:37:46	SELECT * FROM first_table UNION ALL SELECT * FROM second_table	6 row(s) returned
✓ 24	15:38:28	SELECT COUNT(*) FROM (SELECT * FROM first_table UNION ALL SELECT * FROM second_table ...	1 row(s) returned
✓ 25	15:39:06	SELECT * FROM first_table INTERSECT SELECT * FROM second_table	1 row(s) returned
✓ 26	15:39:33	SELECT name FROM first_table INTERSECT SELECT name FROM second_table	1 row(s) returned
✓ 27	15:40:25	SELECT f.* FROM first_table f WHERE NOT EXISTS (SELECT 1 FROM second_table s WHERE s.id...	1 row(s) returned
✓ 28	15:40:52	SELECT f.name FROM first_table f WHERE NOT EXISTS (SELECT 1 FROM second_table s WHER...	1 row(s) returned

